Theory Manual for the MatLab Program
Neutron Spectra Validation and Adjustment

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Chapter 1

Introduction

This manual provides insight of the theory used for the *NSVA* program Neutron Spectrum Validation and Adjustment. It is addressed to the user of this program. An explanation of how to install, run and modify this program is kept separate from this theory part, but available in the User's Manual for *NSVA*.

Neutron radiation has effects on any material. One motivation for the accurate determination of neutron spectra is to improve prediction of neutron displacement damage. The damage caused by this radiation in electronics and structural materials can lead to failure. Therefore it is important to characterize the radiation in such a way that uncertainties in displacement damage effects can be minimized. For more information on this topic please refer to (6).

The applied Neutron Spectra Validation and Adjustment, shortened to *NSVA*, serves the purpose to combine available information about the fluence by the least square method. In an overdetermined system of equations, where each of these equations itself contains uncertainties, the redundancy of information is utilized. A very basic example of this principle is introduced at this point.

Assuming a door, which front surface can be easily determined by it's rectangular geometric form. The surface area, A, can be computed using the height of the door a multiplied by its width b. Let us suppose that the area A can also be measured by an independent method, with result A_m.

$$A = a \cdot b$$

$$A = A_m$$

Each variable on the right can be measured independently. The height a and the width b are measured with the same tool and therefore they are correlated to each other. The uncertainty of each length plays into the calculation.

In the end there are two different equations for the same item. The directly measured surface of the door with its own uncertainty and the calculated result, whose uncertainty can be calculated from the uncertainties in the two length values. Both items can be used to find the best approximation of the *real* value of the surface of the door by the least square method. The same principle, even though on a much larger scale, is used for the neutron spectrum adjustment.

1.1 Organization of this manual

The first Chapter is supposed to give a rough idea about the topic and the purpose of the theory manual. A very much simplified example is given to introduce the least square adjustment method.

The second Chapter introduces general terms, which are also used in related literature. The main focus lays here in the derivation for a special case of the applied least square method. This Chapter is not supposed to teach the fundamentals of this method, but gives insight in the simplifications that cause a loss of generality.

In the third Chapter, the applied and implemented equations of the MatLab program *NSVA* are determined and explained. This chapter should make the code of the program more accessible and understandable to the user. The sequences and explanation of the equation follows the order in which they appear in the program.

The last part of this manual, the Bibliography, lists the references that were the basis for the realization of the MatLab program *NSVA*. The abbreviation *NSVA* is used for the term: *Neutron Spectrum Validation* and *Adjustment*.

Chapter 2

Derivation

This Chapter will not provide a general introduction into least square problems. To review this subject please refer to the textbooks cited in the bibliography (3), (2) and (1), or other related literature. Starting from the previous derivations in (5), another formulation of the described problem will be derived and explained.

2.1 General terms

The neutron fluence of an environment is referred to by the Greek letter Φ. It represents the amount of neutrons passing through a defined area of $1cm^2$. Due to the different velocities with which the neutrons are traveling, the fluence is separated into 89-group energy bins. The faster the neutrons are traveling, the higher their energy. Therefore Φ is a column vector with fluence data ordered from high to low energy. The neutron fluence with highest energy is at the top of the vector, the ones with the lowest at the bottom.

The cross–section is a measure of reaction probability, or the likelihood of interaction between a neutron and an isotope. It has the physical dimension of $1/cm^2$ and is referred to by the Greek letter σ. Since the likelihood of interaction differs for every single group energy bin of the fluence, σ has to have a value for each energy bin. Consequently the set of cross–sections Σ is defined to be a column vector. Furthermore, every isotope and every reaction has a different cross–section. The cross–section vector of a specific reaction i is indicated by Σ_i.

Further variables will be explained in the other chapters since they may differ in terms of notation and scale.

2.2 Review the adjustment formulas

Based on the well known formulation for the reaction probability per atom, the product of the cross–section and the fluence can loosely be referred to as the calculated activity A_c in this program (as in others). Ideally, A_c should be equal to the measured activity per target atom divided by the decay constant.

$$A_c(i) = \Sigma_i^\top \cdot \Phi \tag{2.1}$$

As the reaction probability per atom will be referred to as the activity, every activity i can be computed out of the product of the fluence and the corresponding cross–section i. The physical dimensions of the multiplicands is canceled out, so that the activity is dimensionless. Remembering that the fluence and the cross–section of reaction i both are vectors, equation (2.1) can also be expressed as

$$A_c(i) = \Phi^\top \cdot \Sigma_i \tag{2.2}$$

Equation (2.2) may seem redundant, but will be of further use in Chapter 3.4. Fundamental for the application of least square adjustment is a vector of prior parameters. For the described case of the fluence adjustment, it contains the fluence vector and the cross–section vectors of the involved reactions. The structure of this vector P_0 is stated in equation (2.3)

$$P_0 = \begin{pmatrix} \Phi \\ \underline{\boldsymbol{\Sigma}} \end{pmatrix} \tag{2.3}$$

In this configuration $\underline{\boldsymbol{\Sigma}}$ is also a column vector, containing all cross–section vectors Σ_i of the involved reactions in a specified order.

The uncertainties of a vector containing random variables are represented by a covariance matrix. For some applications, the transformation from a covariance matrix to a correlation matrix is useful. A covariance matrix can be computed out of the actual data the dependency refers to, the standard deviations of the data and the correlation matrix. The direct conversion is shown below.

$$\mathbf{Cov_X} = \mathbf{C} \cdot \mathbf{Cor_X} \cdot \mathbf{C}^\top \tag{2.4}$$

where $\mathbf{C}$ is a diagonal matrix of the standard deviations. The dimension of $\mathbf{C}$ has to be the same as $\mathbf{Cor_X}$. We usually express the standard deviations as fractions or percentages, so $\mathbf{C}$ can be expressed as the product of two diagonal matrices as

$$\mathbf{C} = \begin{bmatrix} x_1 & 0 & \dots & 0 \\ 0 & x_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & x_n \end{bmatrix} \cdot \begin{bmatrix} sd_{x_1} & 0 & \dots & 0 \\ 0 & sd_{x_2} & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & sd_{x_n} \end{bmatrix} \tag{2.5}$$

The least square adjustment formula from (7) for the parameter vector P, whose structure is similar to the vector shown in equation (2.3), is written as[1]

$$\mathsf{P}' = \mathsf{P_0} - \mathbf{Cov}_{P_0} \cdot \mathbf{S}_P \cdot \left(\mathbf{S}_P^\top \mathbf{Cov}_{P_0} \mathbf{S}_P + \mathbf{Cov}_A\right)^{-1} \cdot (\mathsf{A_c} - \mathsf{A_m}) \tag{2.6}$$

In the displayed equation, P_0 is the prior parameter vector and P' is the ad-

[1]The adjustment formula is written directly from the reference. The further use of this formula and it's implementation in the program will show a sign change. This change in sign results from a different order in the subtraction of the activities A_c and A_m.

justed one. $\mathsf{A_c}$ is the calculated vector of reaction probabilities, $\mathsf{A_m}$ the measured vector of reaction probabilities determined from the activity of the reaction products. Since these vectors are obtained differently, they are assumed to be independent from one another. Thus, the inverted term of equation (2.6) can be written as a sum of two independent covariances. The product of the sensitivity matrices of the prior parameter vector $\mathbf{S}_P$ with its covariance matrix $\mathbf{Cov}_{P_0}$ indicates the covariance matrix of the calculated reaction probabilities, while $\mathbf{Cov}_A$ refers to the measured activity probabilities. The two vectors of reaction probabilities, $\mathsf{A_m}$ and $\mathsf{A_c}$, are often referred to as the measured and calculated activities as explained at the beginning of Section 2.2. They actually correspond to the induced activities per target atom divided by their decay constants. The adjustment formula for the measured activity is also drawn from the derivation in (7) and written as[2]

$$\mathsf{A}' = \mathsf{A_m} + \mathbf{Cov}_A \cdot \left(\mathbf{S}_P^\top \mathbf{Cov}_{P_0} \mathbf{S}_P + \mathbf{Cov}_A\right)^{-1} \cdot (\mathsf{A_c} - \mathsf{A_m}) \tag{2.7}$$

For both equations (2.6) and (2.7) the sensitivity matrix $\mathbf{S}_P$, which refers to the parameter vector P_0, contains the cross–section and the fluence vectors. This sensitivity matrix is displayed on the next page as $\mathbf{S}_P$.

[2]In the further implementation of this formula might also show a different sign. This is due to the same reasons as mentioned in (2.6)

$$\mathbf{S}_P = \begin{bmatrix} \Sigma_1 & \Sigma_2 & \dots & \Sigma_n \\ \Phi & 0 & \dots & 0 \\ 0 & \Phi & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & \Phi \end{bmatrix} \tag{2.8}$$

The covariance matrix of the prior parameters includes the covariance matrix of the fluence and the covariance matrices of the cross–section. In a simplified case its entries are only these covariance matrices on the main diagonal as displayed below.

$$\mathbf{Cov}_P = \begin{bmatrix} \mathbf{Cov}_\Phi & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{Cov}_{\Sigma_1} & & \vdots \\ \vdots & & \ddots & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{Cov}_{\Sigma_n} \end{bmatrix} \tag{2.9}$$

In this formulation it is assumed that there are no correlations between the fluences and the cross–sections. Neglecting these cross–correlations is an accepted procedure. Much more questionable is the exemption of the cross correlation terms between the different cross–sections, since they are known to have a non–neglible effect on the adjustment. Unfortunately there were no such data available at the time the program has been written. However, the possibility of implementing these cross correlation terms at a later time has been kept optional. The matrix as it is shown in equation (2.9) is not existent in the program, but a complete matrix with the cross–section correlation matrices on the main diagonal is established in *NSVA* and can be edited to include cross correlation terms. Nevertheless, not considering these dependencies is a loss of generality. The adjustment formulas (2.6) and (2.7) describe the general case, so long as the measurements $\mathsf{A_m}$ are independent of the prior fluences and the cross–sections.

2.3 Implementation of the adjustment formulas

As stated at the end of the previous Chapter 2.2 the reformulation underlies a simplification, which neglects any correlation between the input fluences and the input cross–sections. Considering this simplification the common term of equations (2.6) and (2.7)

$$\mathbf{S}_P^\top \mathbf{Cov}_{P_0} \mathbf{S}_P \tag{2.10}$$

can be extracted to

$$\begin{bmatrix} \Sigma_1^\top & \Phi^\top & 0^\top & \dots \\ \Sigma_2^\top & 0^\top & \Phi^\top & \\ \vdots & \vdots & & \ddots \\ \Sigma_n^\top & 0^\top & & \end{bmatrix} \cdot \begin{bmatrix} \mathbf{Cov}_\Phi & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{Cov}_{\Sigma_1} & & \vdots \\ \vdots & & \ddots & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{Cov}_{\Sigma_n} \end{bmatrix} \cdot \begin{bmatrix} \Sigma_1 & \Sigma_2 & \dots & \Sigma_n \\ \Phi & 0 & \dots & 0 \\ 0 & \Phi & & \\ \vdots & & \ddots & \end{bmatrix}$$

Further expanding (2.10) to a single matrix still maintains a clear pattern. This intermediate step is displayed below.

$$\begin{bmatrix} \Sigma_1^\top \mathbf{Cov}_\Phi \Sigma_1 + \Phi^\top \mathbf{Cov}_{\Sigma_1} \Phi & \Sigma_1^\top \mathbf{Cov}_\Phi \Sigma_2 & \dots & \Sigma_1^\top \mathbf{Cov}_\Phi \Sigma_n \\ \Sigma_2^\top \mathbf{Cov}_\Phi \Sigma_1 & \Sigma_2^\top \mathbf{Cov}_\Phi \Sigma_2 + \Phi^\top \mathbf{Cov}_{\Sigma_2} \Phi & & \Sigma_2^\top \mathbf{Cov}_\Phi \Sigma_n \\ \vdots & & \ddots & \vdots \\ \Sigma_n^\top \mathbf{Cov}_\Phi \Sigma_1 & \Sigma_n^\top \mathbf{Cov}_\Phi \Sigma_2 & \dots & \Sigma_n^\top \mathbf{Cov}_\Phi \Sigma_n + \Phi^\top \mathbf{Cov}_{\Sigma_n} \Phi \end{bmatrix}$$

This single matrix can now be separated into a sum of two matrices. The first matrix, which shall be called $\mathbf{C}_\Phi$, has all terms that include the fluence covariance matrix. The other one, here called $\mathbf{C}_\Sigma$, contains the remaining terms which only appear on the main diagonal. In expanded form these matrices are shown in the following two equations.

$$\mathbf{C}_\Phi = \begin{bmatrix} \Sigma_1^\top \mathbf{Cov}_\Phi \Sigma_1 & \Sigma_1^\top \mathbf{Cov}_\Phi \Sigma_2 & \dots & \Sigma_1^\top \mathbf{Cov}_\Phi \Sigma_n \\ \Sigma_2^\top \mathbf{Cov}_\Phi \Sigma_1 & \Sigma_2^\top \mathbf{Cov}_\Phi \Sigma_2 & & \Sigma_2^\top \mathbf{Cov}_\Phi \Sigma_n \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_n^\top \mathbf{Cov}_\Phi \Sigma_1 & \Sigma_n^\top \mathbf{Cov}_\Phi \Sigma_2 & \dots & \Sigma_n^\top \mathbf{Cov}_\Phi \Sigma_n \end{bmatrix} \tag{2.11}$$

$$\mathbf{C}_\Sigma = \begin{bmatrix} \Phi^\top \mathbf{Cov}_{\Sigma_1} \Phi & 0 & \dots & 0 \\ 0 & \Phi^\top \mathbf{Cov}_{\Sigma_2} \Phi & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & \Phi^\top \mathbf{Cov}_{\Sigma_n} \Phi \end{bmatrix} \tag{2.12}$$

The matrix $\mathbf{C}_\Phi$ now can be expanded to the product of three smaller matrices

$$\mathbf{C}_\Phi = \begin{bmatrix} \Sigma_1^\top \\ \Sigma_2^\top \\ \vdots \\ \Sigma_n^\top \end{bmatrix} \cdot \mathbf{Cov}_\Phi \cdot \begin{bmatrix} \Sigma_1 & \Sigma_2 & \dots & \Sigma_n \end{bmatrix}$$

similarly the matrix $\mathbf{C}_\Sigma$ is expanded to

$$\mathbf{C}_\Sigma = \begin{bmatrix} \Phi^\top & 0^\top & \dots & 0^\top \\ 0^\top & \Phi^\top & & \vdots \\ \vdots & & \ddots & \\ 0^\top & & & \Phi^\top \end{bmatrix} \cdot \begin{bmatrix} \mathbf{Cov}_{\Sigma_1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{Cov}_{\Sigma_2} & & \vdots \\ \vdots & & \ddots & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{Cov}_{\Sigma_n} \end{bmatrix} \cdot \begin{bmatrix} \Phi & 0 & \dots & 0 \\ 0 & \Phi & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & \Phi \end{bmatrix}$$

using this expanded formulation, $\mathbf{C}_\Phi$ and $\mathbf{C}_\Sigma$ may now also be written in a compact form shown in equation (2.13).

$$\mathbf{C}_\Phi = \mathbf{S}_\Phi \cdot \mathbf{Cov}_\Phi \cdot \mathbf{S}_\Phi^\top \qquad \text{and} \qquad \mathbf{C}_\Sigma = \mathbf{S}_\Sigma \cdot \mathbf{Cov}_{\Sigma_{1\dots n}} \cdot \mathbf{S}_\Sigma^\top \tag{2.13}$$

The matrix $\mathbf{S}_\Phi$ shall be called the sensitivity matrix of the fluence, while $\mathbf{S}_\Sigma$ is called the sensitivity matrix of the cross–sections. These matrices are further explained in Chapter 3.4. Summarizing the entire procedure, equation (2.10) at the beginning of this paragraph is equal to the sum of $\mathbf{C}_\Phi$ and $\mathbf{C}_\Sigma$. Substituting equations (2.13) into (2.10) leads to:

$$\mathbf{S}_P^\top \mathbf{Cov}_{P_0} \mathbf{S}_P = \mathbf{S}_\Phi \cdot \mathbf{Cov}_\Phi \cdot \mathbf{S}_\Phi^\top + \mathbf{S}_\Sigma \cdot \mathbf{Cov}_{\Sigma_{1\ldots n}} \cdot \mathbf{S}_\Sigma^\top \tag{2.14}$$

where equation (2.14) can now be substituted into the adjustment formulas (2.6) and (2.7). These reformulated adjustment formulas are displayed below and are implemented in into the MatLab progrm *NSVA*.

$$\mathsf{P}' = \mathsf{P}_0 - \mathbf{Cov}_{P_0} \cdot \mathbf{S}_P \cdot \left(\mathbf{S}_\Phi \cdot \mathbf{Cov}_\Phi \cdot \mathbf{S}_\Phi^\top + \mathbf{S}_\Sigma \cdot \mathbf{Cov}_{\Sigma_{1\ldots n}} \cdot \mathbf{S}_\Sigma^\top + \mathbf{Cov}_A\right)^{-1} \cdot (\mathsf{A}_\mathsf{c} - \mathsf{A}_\mathsf{m}) \tag{2.15}$$

$$\mathsf{A}' = \mathsf{A}_\mathsf{m} + \mathbf{Cov}_A \cdot \left(\mathbf{S}_\Phi \cdot \mathbf{Cov}_\Phi \cdot \mathbf{S}_\Phi^\top + \mathbf{S}_\Sigma \cdot \mathbf{Cov}_{\Sigma_{1\ldots n}} \cdot \mathbf{S}_\Sigma^\top + \mathbf{Cov}_A\right)^{-1} \cdot (\mathsf{A}_\mathsf{c} - \mathsf{A}_\mathsf{m}) \tag{2.16}$$

Chapter 3

Applied Equations in NSVA

3.1 Notation of the Variables

To express the applied equations, this manual utilizes mathematical symbols and Greek letters. Most of these are not accessible in the MatLab code and therefore the expressions here may differ from the ones in the code. However, the descriptions of vectors and matrices are chosen in a way that the equations should be recognizable by the user easily.

The notation of variables leans onto the notation that is used in (5). A detailed list of the variables, their denotation and their representation in the MatLab code is listed in the User's manual.

3.2 Structure of the library data

The program NSVA hosts all necessary data for the calculation which remains constant. Especially to add more data to this library or generally edit the program it is important to understand its structure.

The number of *energy groups* and the number of *reactions* known to the program are essential for the dimensions of many matrices and the data that have to be provided by the user. The number of *energy groups* is expressed as n_g and the number of *reactions* as n_r. The data library of the program refers to a set of fluence data that is separated into 89-group energy bins, ordered from high to low energy. All 34 reactions listed in the interface are represented in the program. Other multi–group structures may be used without modifying the program itself if a different cross–section library is available.

The variable **CrS** found in the sub-folder *library data* contains the cross–section values of all reaction known to the library. In the present implementation here there are 34 reactions in the library. **CrS** has therefore the dimension of n_g rows by n_r columns, (89×34). It is constructed as

$$\mathbf{CrS} = (\Sigma_1, \Sigma_2, \cdots, \Sigma_{n_r})$$

where Σ_i represents the cross–section vector of each reaction i in the library. The variable **SD** is expressed in the same way, where sd_{Σ_i} is the standard deviation in percent to the related cross–section Σ_i.

$$\mathbf{SD} = \left(sd_{\Sigma_1}, sd_{\Sigma_2}, \cdots, sd_{\Sigma_{n_r}}\right)$$

The correlation and covariance matrices of the cross–section are also stored in the data library. The variable containing the cross–sections correlation matrices is called **Cor** while the variable for the cross–section covariance matrices is named **Cova**. Both matrices, **Cor** and **Cova**, are assembled in the same way. Since either variable has to contain square matrices only, a 3 dimensional matrix was introduced. The best way to imagine it, is to simply picture a cube. The first *slice* of this cube contains the first correlation or covariance matrix. For all of the variables listed above it is imperative that the order of the reactions is maintained. As mentioned before, the reactions are ordered after the atomic numbers of their isotopes.

3.3 Structure of the input data

The user has the choice between different formats for the input data. The program is compatible with MatLab Data, MS Excel and regular text files. Depending on the kind of format, the data might get reassembled internally during the process of loading the file. If the program detects a dimensional mismatch in the input data, a warning will provide explicit information which data set causes the error.

3.3.1 Structure of the activity data

A new vector for each environment is introduced to filter the information of the provided activities. This vector is called m_1 for the first environment and m_2 for the second one. Both items are column vectors of the size $(n_r \times 1)$, where n_r is the constant number of reactions in the library. Their entries are restricted to be one or zero, depending if a reaction is included in the calculation or not. One indicates that the reaction is included, zero explains itself. Therefore number of involved reactions for the first environment is:

$$\sum_{i}^{n_r} m(i) = j_1$$

The variable j_2 can be determined similarly.

The number of provided activity data for the input can exceed j_1 for environment one or j_2 for environment two, since distinct reactions can be excluded from the calculation later on. j_1 and j_2 indicate the number of reactions that must be considered in the adjustment. Two other vectors per environment, called m_{E1} and m_{e1}[1], are necessary to create the previously described vector m_1. Both vectors, m_{E1} and m_{e1}, have the same dimension and restrictions as m_1. The vector m_{E1} indicates which response activity data is provided by the user, whereas m_{e1} indicates which reaction shall be excluded from the com-

[1]These vectors, m_{E1} and m_{e1}, are not used any further in this theory manual. However, they are mentioned because they are used in the MatLab program to determine the vector m_1, which is known as $decisionE1$ in the program. It should also give a better idea how the vector m_1 is implemented later on.

putation. Out of these two vectors m_1 is created to actually determine those reactions to be considered in the calculation. The vector m_1 is also used to adjust other necessary data for the computation.

The set of measured activities have to be provided as a column vector. The entries of the activities have to be sorted in the same order as the nuclear data library, which is also the same order as in the window of the GUI. The order is based upon the atomic number, then mass number of the isotopes of the same element and finally the product of the reaction. The same order is required for the column vector of activity standard deviations in percent. The activity correlation matrix has to have a dimension and order according to the provided response activity vector.

Reordering the data might be cumbersome. For this reason the MatLab program includes an independent tool to order the response activity data, the standard deviations and the activity correlation matrix simultaneously. For more information about this tool please refer to the *User's Manual*.

The activity data is structured as

$$A_i = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \qquad sd_{A_i} = \begin{pmatrix} sd_{a_1} \\ sd_{a_2} \\ \vdots \\ sd_{a_n} \end{pmatrix} \qquad \mathbf{Cor}_{A_i} = \begin{bmatrix} 1 & c_{12} & \dots & c_{1n} \\ c_{21} & 1 & \dots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \dots & 1 \end{bmatrix}$$

The correlation matrix of the activities has to be square symmetric and positive semi-definite. After completing the input and starting the computation, the program checks the eigenvalues of the assembled correlation matrix of the activities. In case an eigenvalue of this assembled matrix is smaller than a defined tolerance[2] the program will notify the user by a warning and additionally check all correlation matrices of the activities separately. It will not stop the computation, but continuing the adjustment might produce wrong results.

3.3.2 Structure of the fluence data

The data sets for the neutron environment includes the fluence data itself, the fluence standard deviations and the fluence correlation matrix. The format is somewhat similar to the one of the activities. The fluence data is ordered from high to low energy groups. Since the library data for the cross–section values is based on 89 different energy group bins (n_g), the fluence data must have a compatible dimensions.

$$\Phi_i = \begin{pmatrix} \Phi_1 \\ \Phi_2 \\ \vdots \\ \Phi_{n_g} \end{pmatrix} \qquad sd_{\Phi_i} = \begin{pmatrix} sd_{\Phi_1} \\ sd_{\Phi_2} \\ \vdots \\ sd_{\Phi_{n_g}} \end{pmatrix} \qquad \mathbf{Cor}_{\Phi_i} = \begin{bmatrix} 1 & c_{12} & \dots & c_{1n_g} \\ c_{21} & 1 & \dots & c_{2n_g} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n_g1} & c_{n_g2} & \dots & 1 \end{bmatrix}$$

[2]The tolerance is set manually to a constant value of $eps = -10^{-6}$ to disregard occurrences of numerical errors that might arise. Note that a proper correlation or covariance matrix is positive semi–definite; it should not have negative eigenvalues.

The check described in Section 3.3.1 for the activity data, which determines the required properties of the activity correlation matrix, is also performed on the correlation matrix of the fluence. In case of the simultaneous adjustment of two environments, this check is performed on the constructed correlation matrices of the fluences including both correlation matrices and their cross–correlation matrix.

3.4 Structure of internally created variables

After completing the input of the required data the program starts creating new variables for the adjustment with execution of the *compute* button. Recalling the formulations of the calculated activity A_c from Chapter 2.1, it is known that the sensitivity of the fluence corresponds to the differentiation of equation (2.1) with respect to the fluence. The sensitivity matrix of the fluence can be represented as

$$\mathbf{S}_{\Phi} = \begin{bmatrix} \Sigma_1^{\top} & \rightarrow \\ \Sigma_2^{\top} & \rightarrow \\ \vdots & \vdots \\ \Sigma_{n_r}^{\top} & \rightarrow \end{bmatrix} \tag{3.1}$$

where S_{Φ} contains all transposed cross–section vectors Σ which causes the dimension to be [n_r by n_g]. Similarly, the sensitivity matrix of the cross–sections can easily be determined by the differentiation of equation (2.2) with respect to the cross–sections

$$\mathbf{S}_\Sigma = \begin{bmatrix} \Phi^\top \rightarrow & 0^\top \rightarrow & \dots & 0^\top \\ 0^\top \rightarrow & \Phi^\top \rightarrow & & \vdots \\ \vdots & & \ddots & 0^\top \\ 0^\top \rightarrow & \dots & 0^\top & \Phi^\top \end{bmatrix} \tag{3.2}$$

so that S_Σ is now a $[n_r$ by $(n_r \cdot n_g)]$ matrix containing the same fluence vector over and over again.

For the simultaneous adjustment of two environments, equation (2.1) must be reformulated

$$A_c = \begin{bmatrix} \mathbf{S}_{\Phi_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_{\Phi_2} \end{bmatrix} \cdot \begin{pmatrix} \Phi_1 \\ \Phi_2 \end{pmatrix} \qquad \text{where } A_c = \begin{pmatrix} A_{c1} \\ A_{c2} \end{pmatrix} \tag{3.3}$$

This matrix vector expression calculates the reaction probabilities of both environments in a single equation independent from one another. It is also possible to rewrite an expression for the combined calculated activity vector by reformulating equation (2.2) as shown on the next page.

$$A_c = \begin{bmatrix} \mathbf{S}_{\Sigma_1} \\ \mathbf{S}_{\Sigma_2} \end{bmatrix} \cdot \begin{pmatrix} \Sigma_1 \\ \Sigma_2 \\ \vdots \\ \Sigma_{n_r} \end{pmatrix} \tag{3.4}$$

The assembled sensitivity matrices of the previous two equations are of further use for the adjustment and are therefore stored as new matrices

$$\mathbf{S}_{2\Phi} = \begin{bmatrix} \mathbf{S}_{\Phi_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_{\Phi_2} \end{bmatrix} \qquad \text{and} \qquad \mathbf{S}_{2\Sigma} = \begin{bmatrix} \mathbf{S}_{\Sigma_1} \\ \mathbf{S}_{\Sigma_2} \end{bmatrix} \tag{3.5}$$

For computational reasons equation (3.3) is preferred to calculate the reaction probabilities, since the number of multiplications is much smaller compared to equation (3.4). In any case, the sensitivity matrices have to be adjusted in such a way that only those activities are calculated that are selected by the user to be considered in the calculation for the adjustment. Therefore the vector m_1, which got introduced at the beginning of chapter 3.3.1, is further utilized. As it has been stated in chapter 3.2, the library data is ordered in the same way the reactions are listed in the interface. The provided activity data has been in ordered in the same way. Therefore vector m_1 indicates not only how many reactions are involved, but also which ones these are in the data library. According to the reactions that have been excluded from the computation, the corresponding rows of both sensitivity matrices, $\mathbf{S}_\Phi$ and $\mathbf{S}_\Sigma$, are deleted. This procedure is executed for both environments, one and two.

The resulting dimension of the sensitivity matrix $\mathbf{S}_{\Phi_1}$ for environment one now changes from n_r rows and n_g columns to j_1 rows and still n_g columns, since only the rows have been deleted. The number of rows from the sensitivity matrix of the cross–sections, $\mathbf{S}_{\Sigma_1}$, changes similarly from n_r to j_1, while the number of columns remains the same $(n_r \cdot n_g)$. This procedure is identically the same for the second environment. The complete procedure results the combined sensitivity matrix of the fluences to be of the dimension $[(j_1 + j_2)$ by $(2 \cdot n_g)]$ and the combined sensitivity matrix of the cross–sections to be $[(j_1 + j_2)$ by $(n_r \cdot n_g)]$.

Another adjustment needs to be done for the provided activity data. The excluded response activities and their standard deviations must be removed, which causes the size of their vectors to decrease to $[j_1$ by $1]$. The provided activity correlation matrix is reduced by those rows and columns that referred to the excluded reactions. The result is again a square matrix with a dimension of j_1. The discussed items were shown at the end of chapter 3.3.1. As said before, this procedure is also executed similarly for both environments, one and two.

Using the least square adjustment formulas, shown in equations 2.15 and 2.16, for two fields requires a vector of prior parameters including both fluences and the cross–sections. For the described case of two environments the prior parameter vector has a length of $[((2 + n_r) \cdot n_g)$ by $1]$.

$$P_0 = \begin{pmatrix} \Phi_1 \\ \Phi_2 \\ \Sigma_1 \\ \Sigma_2 \\ \vdots \\ \Sigma_{n_r} \end{pmatrix}$$

However, the adjustments in the cross–sections are not required in the output and can be excluded from the calculation. The new introduced vector *phi* reduces the vector of prior parameters.

$$\Phi_{12} = \begin{pmatrix} \Phi_1 \\ \Phi_2 \end{pmatrix} \tag{3.6}$$

To be able to use the adjustment formula, displayed in equation 2.15, the covariance matrix and the sensitivity matrix of the prior parameter vector is now replaced by a covariance matrix and a sensitivity matrix according to the new vector Φ_{12}. The required sensitivity matrices for this vector have already been assembled as displayed in equation (3.5). The corresponding covariance matrix is established by the correlation matrices and the standard deviations of both fluences. Since the prior fluences clculated for two fields might be correlated with each other, the user has to provide a cross correlation matrix for the fluences. Both fluence correlation matrices are square matrices of (89×89) and their cross correlation matrix has the same dimension. If i

indicates the rows and j the columns of the matrices, the element on position (i, j) of the cross correlation matrix indicates the correlation of the i^{th} fluence energy group of the first environment with the j^{th} fluence energy group of the second environment. The combined correlation matrix is written as

$$\mathbf{Cor}_{\Phi_{12}} = \begin{bmatrix} \mathbf{Cor}_{\Phi_1} & \mathbf{CCF} \\ \mathbf{CCF}^\top & \mathbf{Cor}_{\Phi_2} \end{bmatrix} \tag{3.7}$$

where $\mathbf{Cor}_{\Phi_{12}}$ has a dimension of $(2 \cdot n_g \times 2 \cdot n_g)$. To obtain the combined covariance matrix for this combined correlation matrix, it is possible to use the same procedure that is shown in equations (2.4) and (2.5). Note that the cross–correlation matrix **CCF** is not necessarily symmetric.

Similar to the combined fluence correlation matrix, the activity correlation matrix is assembled as

$$\mathbf{Cor}_{A_{12}} = \begin{bmatrix} \mathbf{Cor}_{A_1} & \mathbf{CCA} \\ \mathbf{CCA}^\top & \mathbf{Cor}_{A_2} \end{bmatrix} \tag{3.8}$$

where the activity cross correlation matrix, **CCA**, is provided by the user with the same number of rows as there are measured activities for the first environment and the same number of columns as there are measured activities for environment two. The cross correlation matrix of the activities is adjusted automatically if there are distinct reactions excluded from the computation. The cancellation of distinct rows and columns is also determined by the vectors

m_1 and m_2. Thus, the provided cross correlation matrix must be in a specific order to match the previously reordered activity correlation matrices. The MatLab program includes another external tool to switch rows and columns for this matrix. Further instructions about using this tool is provided in the *User's Manual*.

By taking all new introduced variables into account for the simultaneous adjustment of two environments, the adjustment formula for the fluence vector Φ_{12} can be written as

$$\Phi'_{12} = \Phi_{12} + \mathbf{Cov}_{\Phi_{12}} \cdot \mathbf{S}^{\top}_{\Phi_{12}} \cdot \mathbf{h} \cdot (\mathsf{A}_{\mathsf{m}} - \mathsf{A}_{\mathsf{c}}) \tag{3.9}$$

$$\text{where} \qquad \mathbf{h} = \left(\mathbf{S}_{\Phi_{12}} \cdot \mathbf{Cov}_{\Phi_{12}} \cdot \mathbf{S}^{\top}_{\Phi_{12}} + \mathbf{S}_{\Sigma_{12}} \cdot \mathbf{Cov}_{\Sigma_{1\ldots n}} \cdot \mathbf{S}^{\top}_{\Sigma_{12}} + \mathbf{Cov}_{A_{12}}\right)^{-1}$$

Despite the described changes, the terminology of the adjustment formula stays the same. The sign change is due to the different order in the difference of measured and calculated activity probabilities. The sensitivity and the covariance matrices of the cross–sections are still considered in the adjustment. Even after replacing $\mathbf{Cov}_{P_0}$ and $\mathbf{S}_P$ by $\mathbf{Cov}_{\Phi_{12}}$ and $\mathbf{S}^{\top}_{\Phi_{12}}$[3], the inverted component $\mathbf{h}$ remains unchanged. The correlation matrices of the fluences are not adjusted directly. These matrices can be obtained from the adjusted

[3]The sensitivity matrix of the vector of prior parameters is replaced by the *transposed* sensitivity matrix of the fluences just because $\mathbf{S}_{\Phi_{12}}$ is defined differently as it can be seen in equation (2.14)

covariance matrices, which are calculated as

$$\mathbf{Cov}'_{\Phi_{12}} = \mathbf{Cov}_{\Phi_{12}} - \mathbf{Cov}_{\Phi_{12}} \cdot \mathbf{S}^{\top}_{\Phi_{12}} \cdot \mathbf{h} \cdot \mathbf{S}_{\Phi_{12}} \tag{3.10}$$

Since the adjusted covariance matrix of the fluences contains adjusted values for the standard deviations, the conversion from covariance to correlation matrix is an inversion of the procedure shown in equations (2.4) and (2.5). The main diagonal of the covariance matrix contains the variances of the fluences. The variance is the square of the standard deviations. The off diagonal terms indicate the covariances within the fluences. Thus the covariance matrix can be written as

$$\mathbf{Cov}_{\mathbf{X}} = \begin{bmatrix} \sigma_1^2 & c_{12} & \cdots \\ c_{21} & \sigma_2^2 & \\ \vdots & & \ddots \end{bmatrix} \tag{3.11}$$

The definition of the covariance matrix requires that the opposing covariances must be equal, so that $c_{12} = c_{21}$. Extracting the variances from the covariance matrix leads to the adjusted correlation matrix by multiplying the adjusted standard deviations from both sides

$$\mathbf{Cor}_{\mathbf{X}} = \mathbf{C} \cdot \mathbf{Cov}_{\mathbf{X}} \cdot \mathbf{C} = \begin{bmatrix} 1 & \frac{c_{12}}{\sigma_1 \sigma_2} & \cdots \\ \frac{c_{21}}{\sigma_1 \sigma_2} & 1 & \\ \vdots & & \ddots \end{bmatrix} \tag{3.12}$$

where $\mathbf{C}$ from equation (3.12) is denoted as $\mathbf{C} = \begin{bmatrix} \frac{1}{\sigma_1} & 0 & \cdots \\ 0 & \frac{1}{\sigma_2} & \\ \vdots & & \ddots \end{bmatrix}$

The adjusted reaction probabilities do not require any more variables and can be calculated as

$$\mathsf{A}' = \mathsf{A_m} - \mathbf{Cov}_A \cdot \mathbf{h} \cdot (\mathsf{A_m} - \mathsf{A_c}) \tag{3.13}$$

where $\mathbf{h}$ is the same as it is for equation (3.9). Similarly, for the activity correlation matrix, the computation of the activity covariance matrix is required at first. The adjusted activity covariance matrix is written as

$$\mathbf{Cov}'_A = \mathbf{Cov}_A - \mathbf{Cov}_A \cdot \mathbf{h} \cdot \mathbf{Cov}_A \tag{3.14}$$

where $\mathbf{h}$ is already described in equation (3.9). To obtain the adjusted activity correlation matrix, the same procedure that is applied for the fluences has to be performed here. The correlation matrix of the activities has the same structure as it is shown in equation (3.11) and the adjusted activity correlation matrix is obtained by utilizing equation (3.12).

The program is designed to save prior and adjusted data of the fluences and the reaction probabilities. Most additional data, like the adjusted standard deviation can be extracted from adjusted covariance matrices directly. Other data, like the adjustment or the adjustment percentages are calculated from the prior and adjusted data. It might appear that the uncertainties for some items have increased, which is not true. The adjusted deviations are equal or smaller than the prior values, e.g. (4). However, the standard deviation percentage might increase compared to its prior value. This can occur when an item is adjusted to a lower value, which then increases the value of the standard deviation in units of percent. The actual standard deviation associated with the percentage is always equal or lower than its prior value. The adjustment using the least square method is applied four times, for equation (3.9), (3.10), (3.13) and (3.14).

All other resulting data, except the calculation of Chi–squared, is based on these equations. The statistical value Chi–squared is calculated as

$$\chi^2 = (\mathsf{A_m} - \mathsf{A_c})^\top \cdot \mathbf{h} \cdot (\mathsf{A_m} - \mathsf{A_c}) \tag{3.15}$$

Chi–squared (χ^2) refers to a statistic, following a known distribution function (pdf) whose values indicate the degree of difference between random variables whose expectation value is the same. The expected value of χ^2 is equal to the number of degrees of freedom of a system. In our problem, the number of degrees of freedom is equal to the number of involved reactions ($j_1 + j_2$ for the simultaneous adjustment of two environments) less the number of reactions used for arbitrary normalization. The probability that χ^2 per degree of freedom is greater than a specified value is given in tables for the χ^2 distribution function, e.g. (1).

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